Single agent xN vs MAPPO vs IPPO: multi-robot coverage

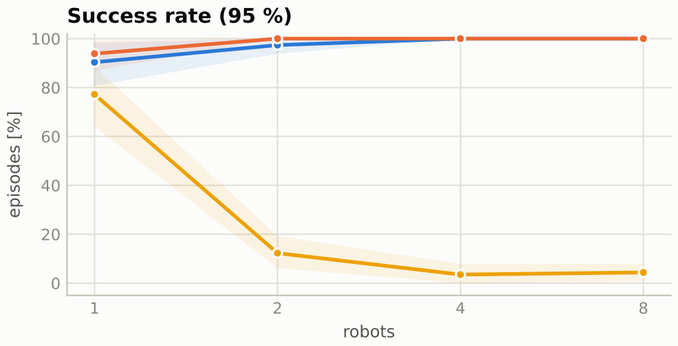
Same episodes for every policy (maps, start poses, seed), deterministic actions. Lines: mean over maps, band: 95 % bootstrap CI over maps.
Without flip_when_stuck. Team efficiency relative to Single-agent xN with 1 robot on the same map.

Single-agent xN MAPPO IPPO

paper

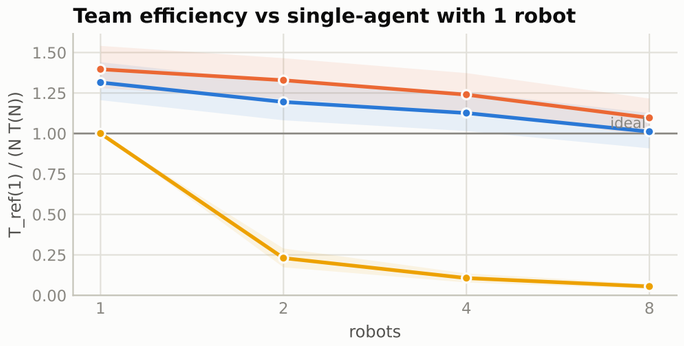
38 maps x 3 episodes, goal 95 % coverage, limit 333 min

Success rate (95 %)

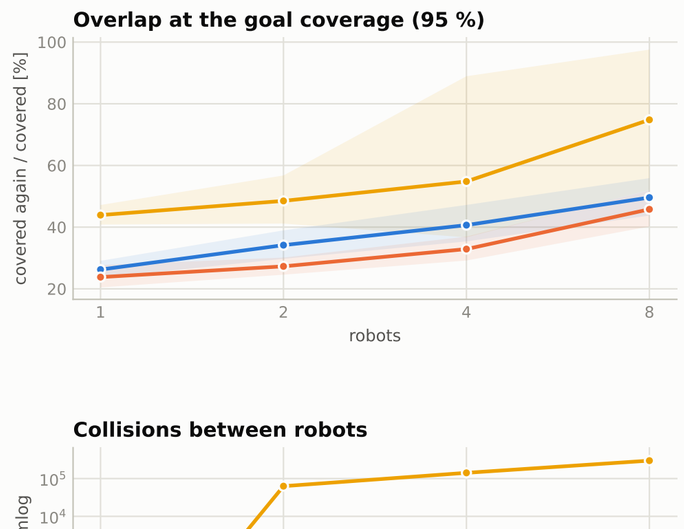


Time to the goal coverage (95 %)

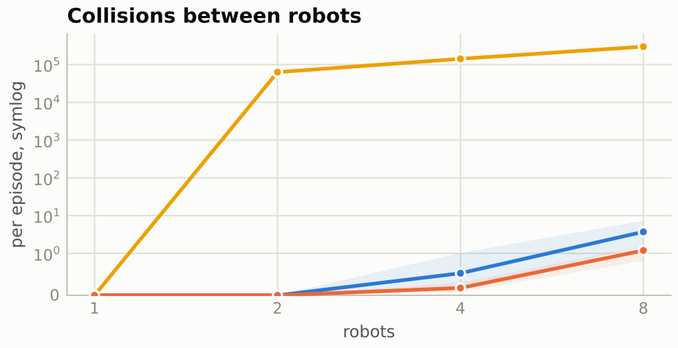
Team efficiency vs single-agent with 1 robot



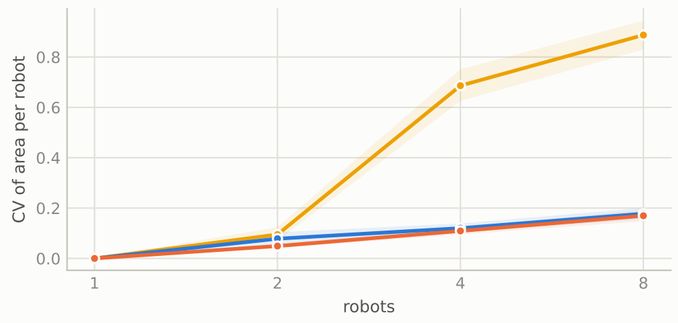
Overlap at the goal coverage (95 %)



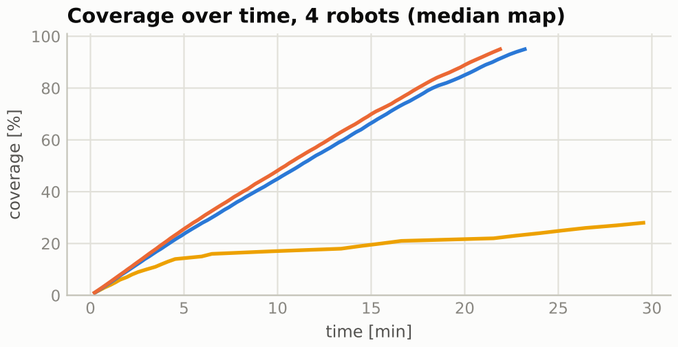
Collisions between robots



Workload imbalance



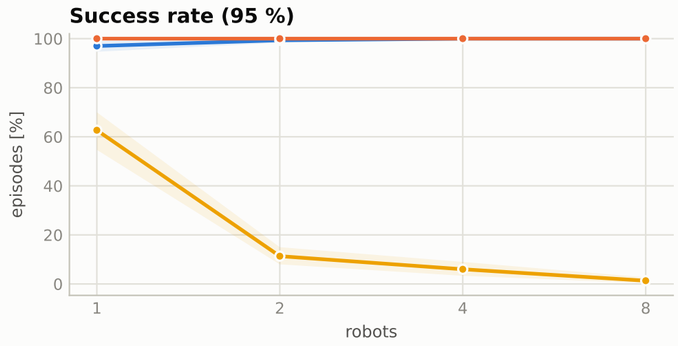
Coverage over time, 4 robots (median map)



test_large

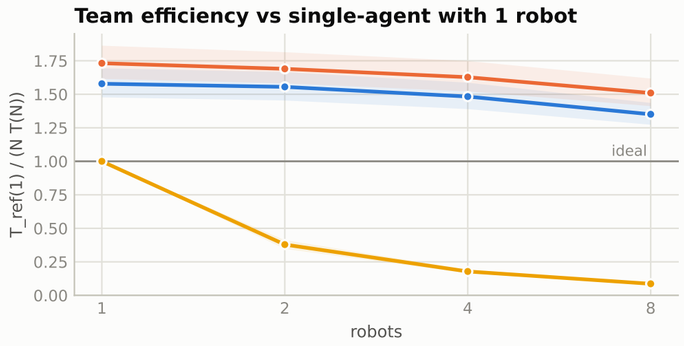
100 maps x 3 episodes, goal 95 % coverage, limit 250 min

Success rate (95 %)

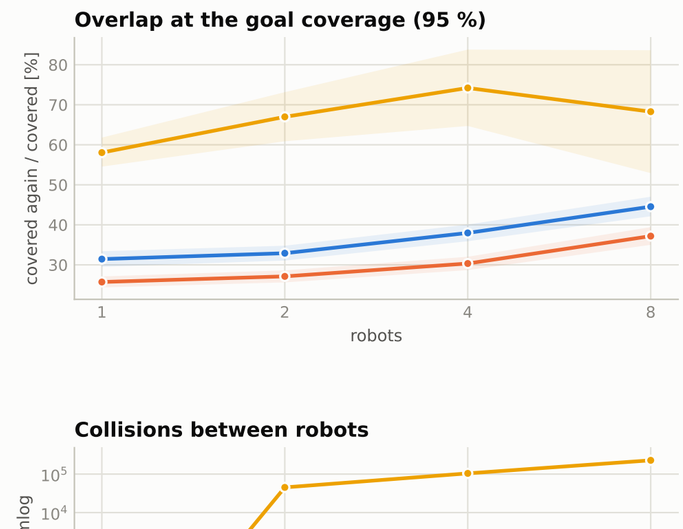


Time to the goal coverage (95 %)

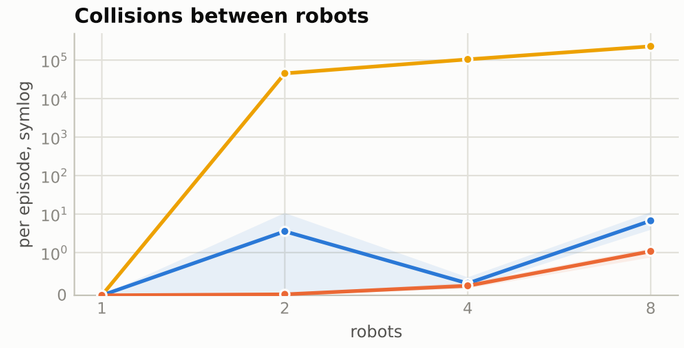
Team efficiency vs single-agent with 1 robot



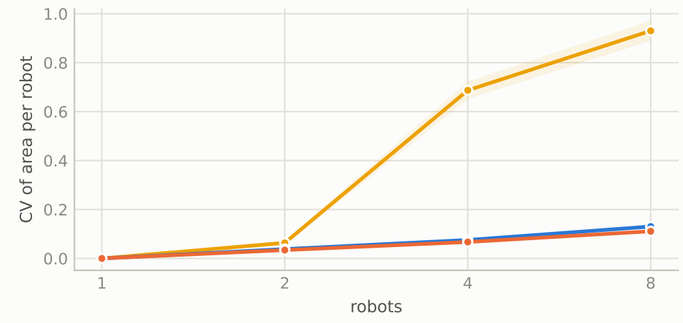
Overlap at the goal coverage (95 %)



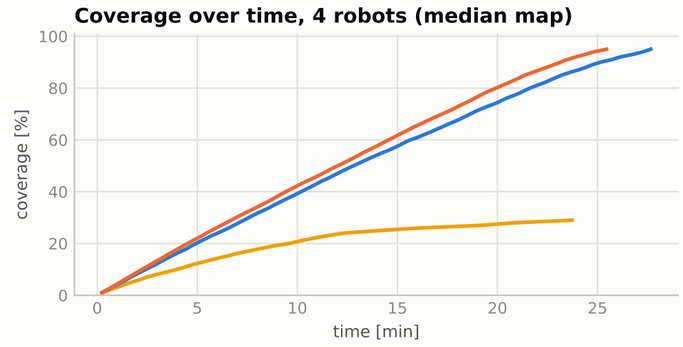
Collisions between robots



Workload imbalance



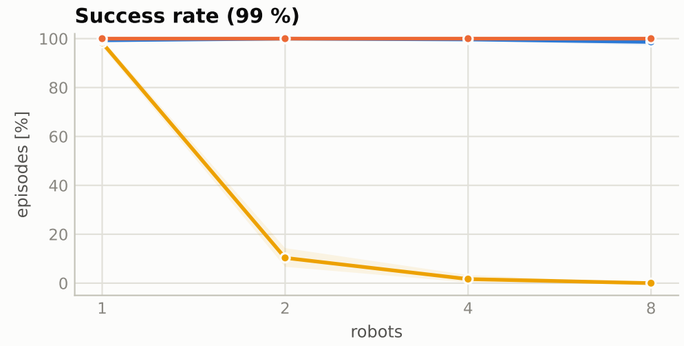
Coverage over time, 4 robots (median map)



test_small

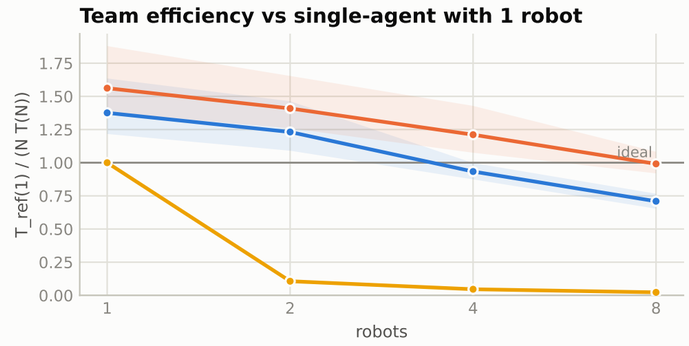
100 maps x 3 episodes, goal 99 % coverage, limit 67 min

Success rate (99 %)

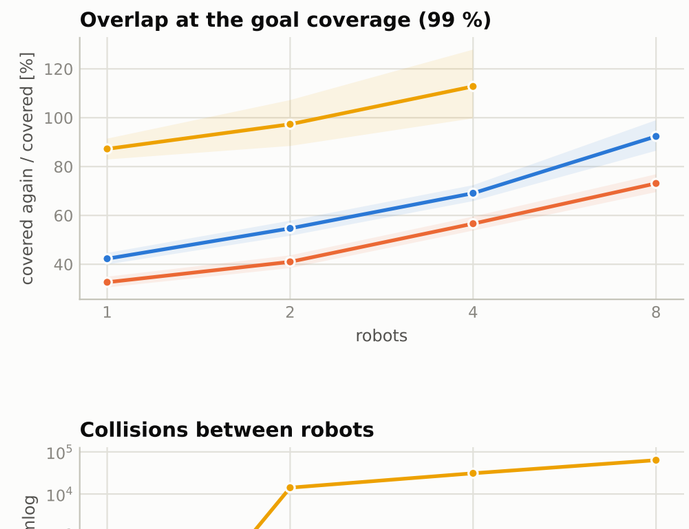


Time to the goal coverage (99 %)

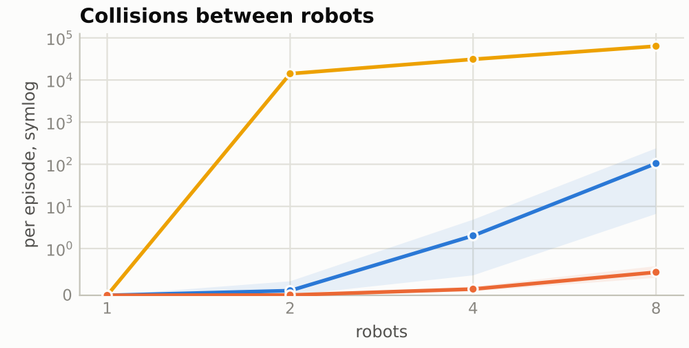
Team efficiency vs single-agent with 1 robot



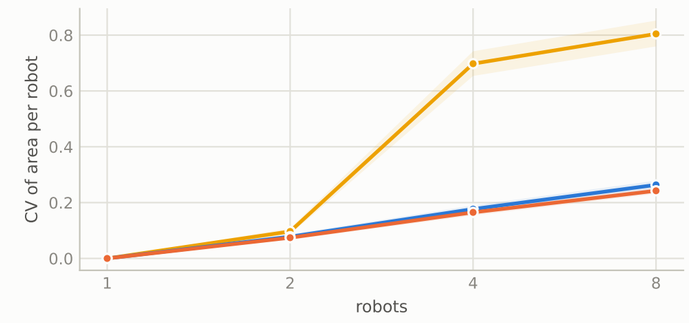
Overlap at the goal coverage (99 %)



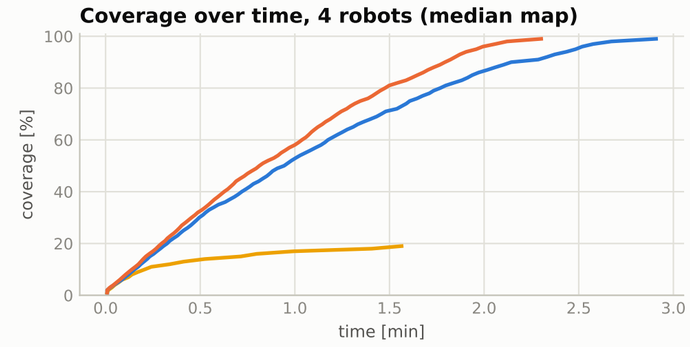
Collisions between robots



Workload imbalance



Coverage over time, 4 robots (median map)



Mean over all map sets and team sizes (N = 1, 2, 4, 8), each (map set, N) weighs the same; best of each column highlighted

	Success %	Time to goal min	Team efficiency T_ref(1) / (N T(N))	Overlap %	Robot collisions per episode	Imbalance CV of area	Path length m per robot
Single-agent xN	24.1	181.4	0.351	71.5	82410.3	0.412	**316**
MAPPO	98.5	37.8	1.239	46.3	10.1	0.095	414
IPPO	**99.5**	**34.0**	**1.399**	**37.8**	**0.3**	**0.085**	386